

Transition Analysis Toolkit

A Reproducible Analysis of Student Transition Pathways

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1 Introduction

The transition from secondary school to first-year university mathematics remains a significant challenge for many students entering science, engineering, and related disciplines. Differences in prior mathematical preparation, curriculum pathways, and academic readiness contribute to wide variation in student success during the first year of university study. For institutions, understanding these transition pathways is important not only for improving student outcomes, but also for informing curriculum design, targeted support programs, and evidence-based educational practice. This study is motivated by the need to better understand the factors associated with successful transition into first-year university mathematics and to provide insights that can support both students and educators.

Over the past two decades, considerable research has examined factors associated with success in first-year university mathematics, including prior academic achievement, secondary school mathematics background, demographic characteristics, and institutional measures of academic preparedness. More recently, advances in educational data analytics and machine learning have enabled increasingly sophisticated predictive models of student performance. While these approaches have improved predictive capability, there remains a need for analyses that not only identify influential predictors but also interpret their relative importance in ways that are meaningful for educational decision-making. This study addresses that need through a transparent and reproducible analytical framework that combines predictive modelling with educational interpretation to better understand transition pathways into first-year university mathematics.

This paper investigates factors associated with successful transition into a first-stage undergraduate mathematics subject using a reproducible analytical workflow that integrates statistical analysis, machine learning, independent verification, and automated reporting. Rather than relying on a single predictive model, multiple tuned decision tree and random forest models are compared to evaluate both predictive performance and the consistency of feature importance across modelling approaches. By combining robust analytical methods with educational interpretation, the study aims not only to identify predictors of student success but also to provide practical insights that may inform curriculum design, transition support, and future research in first-year mathematics education.

1.1 Research Questions

1. Which factors are the strongest predictors of success in first-year university mathematics?
2. How consistent are predictor importance rankings across multiple machine learning models?
3. What educational insights can be drawn from the tuned models?

2 Methodology

This study adopted a reproducible analytical workflow to investigate factors associated with successful transition into first-year university mathematics. The workflow was designed to ensure that each stage of the analysis, from data preparation through statistical modelling, independent verification, and report generation, could be reproduced from the original research dataset using an automated build process. This approach emphasises transparency, repeatability, and traceability, allowing both analytical results and publication outputs to be generated consistently from the same underlying evidence.

2.1 Research Workflow

This study followed a reproducible analytical workflow consisting of data preparation, statistical analysis, machine learning model development, independent verification using R, and automated generation of publication-ready tables, figures, and manuscript components. Each stage formed part of an integrated build process, ensuring that analytical outputs could be reproduced directly from the underlying research dataset.

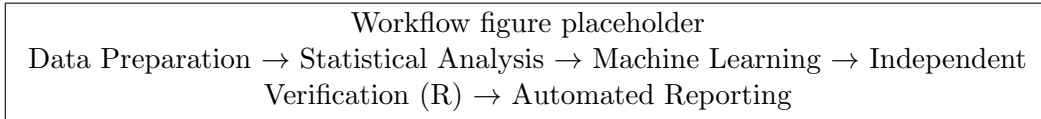


Figure 1: Overview of the reproducible analytical workflow used in this study.

2.2 Research Dataset

The study used de-identified institutional data from students enrolled in a first-stage undergraduate mathematics subject offered during the Autumn 2026 teaching session. The dataset was prepared specifically for research purposes using an approved de-identification process that removed all personally identifying information while preserving the academic and demographic characteristics required for analysis. The resulting dataset provided the foundation for all subsequent statistical analyses, machine learning models, independent verification procedures, and automated reporting undertaken in this study.

2.3 Variables

The analytical dataset comprised variables describing students' prior academic preparation, previous university mathematics experience, demographic characteristics, and current academic outcomes. Variables were selected to reflect factors commonly associated with transition into first-year university mathematics and were derived from institutional records following the approved research data preparation process. A complete description of each variable is provided in Table 1.

Table 1: Summary of variables included in the analytical dataset.

Variable	Description
<i>Placeholder for variable summary table</i>	

2.4 Statistical Analysis

Statistical analyses were undertaken to summarise the characteristics of the research dataset and to examine relationships between student background variables and academic outcomes. Descriptive statistics were generated to provide an overview of the cohort, while inferential analyses supported comparison of predictor variables prior to the development of machine learning models. These analyses established a statistical baseline against which the predictive models could subsequently be interpreted.

2.5 Machine Learning Models

Machine learning models were developed to investigate the relative importance of factors associated with successful transition into first-year university mathematics. Decision tree and random forest classifiers were selected because they provide both competitive predictive performance and interpretable measures of feature importance. Hyperparameter tuning and model

evaluation were undertaken using reproducible workflows to ensure consistent comparison across modelling approaches.

2.6 Independent Verification

To strengthen confidence in the analytical results, an independent verification pipeline was implemented using the R statistical computing environment. Key statistical summaries and model outputs were reproduced independently of the primary Python implementation to confirm the consistency and reliability of the analytical workflow. This independent verification formed an additional quality assurance stage prior to the generation of publication outputs.

2.7 Automated Reporting

The final stage of the analytical workflow automatically generated publication-ready tables, figures, model summaries, and manuscript components directly from the verified analytical outputs. By integrating analysis and reporting within a single reproducible workflow, the potential for manual transcription errors was reduced while ensuring that all reported results remained synchronised with the underlying research data and analyses.

The methodology described above established a transparent and reproducible analytical framework for investigating factors associated with successful transition into first-year university mathematics. The following section presents the outcomes of this workflow, beginning with an evaluation of predictive model performance before examining the relative importance and educational interpretation of the identified predictors.

3 Results

This section presents the findings generated by the reproducible analytical workflow described in the preceding methodology. Results are presented in four stages. First, the predictive performance of the machine learning models is evaluated. Second, the relative importance of predictor variables is examined. Third, the consistency of feature rankings across modelling approaches is assessed. Finally, the educational significance of these findings is interpreted in the context of student transition into first-year university mathematics.

3.1 Model Performance

Table 2: ROC AUC comparison for cross-validated Decision Tree and Random Forest models.

Model	CV ROC AUC	Test ROC AUC	Test Rows
Decision Tree — With Risk	0.7072	0.7664	295
Random Forest — Without Risk	0.7101	0.7603	295
Random Forest — With Risk	0.7095	0.7599	295
Decision Tree — Without Risk	0.6994	0.7557	295

3.2 Feature Importance

Rank	Research theme	Mean importance
1	Secondary mathematics preparation	0.2755
2	Composite risk measure	0.2381
3	Preparatory mathematics	0.2320
4	Previous university performance	0.1090
5	Secondary school achievement	0.1058
6	Student characteristics	0.0395

3.3 Feature Rankings

Rank	Feature	Models	Mean importance
1	Composite risk index	2	0.4762
2	Prior preparatory mathematics	4	0.1694
3	Secondary mathematics: Other	4	0.1019
4	Secondary mathematics: Standard	4	0.0950
5	Previous attempt at current subject	4	0.0693
6	Previous preparatory mathematics pass	4	0.0532
7	Previous failure in current subject	4	0.0397
8	International student	4	0.0395
9	Secondary school mark below 70	4	0.0394
10	Secondary mathematics: Extension 1	4	0.0348
11	Secondary school mark 90 or above	4	0.0288
12	Secondary mathematics: Extension 2	4	0.0257
13	Secondary school mark 80–89	4	0.0228
14	Secondary mathematics: Advanced	4	0.0182
15	Secondary school mark 70–79	4	0.0148
16	Previous preparatory mathematics failure	4	0.0094

3.4 Model Interpretation

Across the tuned tree-based models, secondary mathematics preparation emerged as the most influential predictor group (mean importance = 0.276). This was followed by the composite risk measure (0.238) and preparatory mathematics (0.232). Together, these three groups accounted for 74.6% of the average feature importance across the tuned models. Secondary mathematics preparation remained influential across all four models, whereas the composite risk measure contributed strongly but appeared in two of the four models. In contrast, student characteristics made the smallest average contribution. This suggests that prior secondary mathematics preparation was the strongest and most consistent predictor of success across the tuned models.

4 Discussion

The results of this study provide new insights into the factors associated with successful transition into first-stage university mathematics. By combining reproducible analytical methods with interpretable machine learning models, the analysis extends beyond prediction alone to examine the relative importance and consistency of the variables associated with student success. This discussion considers these findings in the context of previous research, reflects on their educational significance, and identifies implications for both institutional practice and future research.

As demonstrated in Section 3.2, secondary mathematics preparation emerged as the strongest and most influential predictor group across the tuned models, with a mean feature importance of 0.276. Composite risk (0.238) and preparatory mathematics (0.232) were the next most influential predictor groups. Collectively, these three groups accounted for 74.6% of the average feature importance, indicating that measures of prior mathematical preparation dominated the predictive performance of the evaluated models. Furthermore, the consistency of the feature rankings presented in Section 3.3 showed that secondary mathematics preparation remained influential across all four tuned models, whereas the composite risk measure contributed strongly but appeared in only two models. Together, these findings provide robust evidence that students' secondary mathematical preparation was the strongest and most consistent predictor of success in this cohort.

5 Conclusion

This study demonstrated that combining traditional statistical analysis with interpretable machine learning provides a transparent and reproducible framework for investigating student transition into first-year university mathematics. Across the evaluated models, measures of prior mathematical preparation consistently emerged as the strongest predictors of academic success, with secondary mathematics preparation contributing the greatest overall predictive importance. These findings reinforce the importance of discipline-specific educational background in understanding student transition and illustrate how multiple analytical approaches can be integrated to provide both predictive performance and educational interpretation.

The study also demonstrates the value of an integrated and reproducible analytical workflow for educational research. Statistical modelling, machine learning, independent verification, automated reporting, and publication-ready document generation were combined within a single research framework, ensuring that all reported results were derived directly from the underlying analysis. This approach reduces opportunities for transcription errors, improves transparency, and provides a reusable foundation for future studies investigating student transition, progression, and academic success across comparable educational contexts.

The framework developed in this study provides a foundation for future investigations of student transition using larger cohorts, additional academic disciplines, and longitudinal datasets. While the present case study focused on a single first-year mathematics subject, the methodology is readily extensible to other educational contexts where transparent interpretation of predictive models is essential. Future work will investigate broader institutional datasets, evaluate additional modelling approaches, and further develop the automated research workflow to support reproducible educational analytics and evidence-informed decision making.

A A Brief Primer on Interpretable Machine Learning for Educational Research

This appendix provides a concise introduction to the principal machine learning concepts used throughout this study. It is intended for readers with a background in educational research who may be less familiar with predictive modelling. The appendix is explanatory rather than comprehensive and is designed to support interpretation of the methods and results presented in the main paper.

Part I — Machine Learning Foundations

A.1 Why Use Machine Learning in Educational Research?

Educational researchers have traditionally used statistical models to examine relationships between variables and explain differences in student outcomes. Machine learning complements these approaches by placing greater emphasis on prediction. Rather than beginning with a small set of predefined relationships, machine learning models can identify patterns across many variables and evaluate how well those patterns generalise to new data.

In educational settings, this can be useful when the research aim is not only to explain why students succeed or struggle, but also to identify which combinations of prior preparation, academic history, and student characteristics are most informative. When used carefully, machine learning can therefore support early identification, model comparison, and evidence-informed intervention without replacing the need for educational interpretation.

A.2 Prediction and Explanation

Statistical modelling and machine learning often overlap, but they usually emphasise different questions. Statistical models commonly focus on estimating relationships, testing hypotheses, and explaining how particular variables are associated with an outcome. Machine learning models place greater emphasis on predictive performance: how accurately a model can classify or estimate outcomes for observations that were not used during model fitting.

Neither emphasis is inherently superior. The appropriate approach depends on the research question. In educational research, explanation is essential when the aim is to understand mechanisms, evaluate theory, or estimate the effect of an intervention. Prediction is valuable when the aim is to identify students who may require support, compare alternative models, or detect complex patterns that may not be captured by a simpler predefined relationship. Used together, statistical and machine learning approaches can provide complementary evidence.

Part II — Tree-Based Machine Learning

A.3 Decision Trees

A decision tree is one of the simplest and most interpretable machine learning models. It predicts an outcome by repeatedly dividing the data into smaller groups according to decision rules based on the available predictor variables. Each split is chosen to maximise the separation between observations with different outcomes, producing a tree-like structure that can be followed from the root node to a final prediction. Figure 2 illustrates the basic structure of a decision tree, showing how successive decision rules partition observations into increasingly homogeneous groups.

One of the principal advantages of decision trees is their transparency. Every prediction can be traced through a sequence of explicit decisions, allowing researchers to understand why a particular outcome was reached. For this reason, decision trees are widely used as both predictive

models and explanatory tools, particularly in educational research where interpretability is often as important as predictive accuracy.

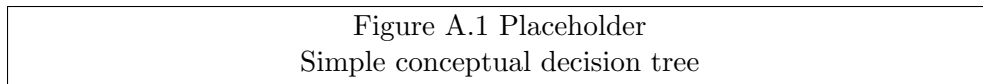


Figure 2: A simplified decision tree illustrating how successive decision rules partition observations into increasingly homogeneous groups.

A.4 Random Forests

Although decision trees are easy to interpret, they can be sensitive to small changes in the training data. A different sample of observations may produce a noticeably different tree, reducing the stability of predictions. Random forests address this limitation by combining the predictions of many decision trees rather than relying on a single model.

Each tree in a random forest is trained using a different bootstrap sample of the available data, and at each decision point only a random subset of predictor variables is considered. This deliberate introduction of randomness produces a diverse collection of trees whose combined prediction is typically more accurate and more robust than that of an individual tree. The final prediction is obtained by aggregating the predictions from all trees, commonly through majority voting for classification problems or averaging for regression problems. The overall concept is illustrated in Figure 3, where the predictions of multiple decision trees are aggregated to produce a single, more stable prediction.

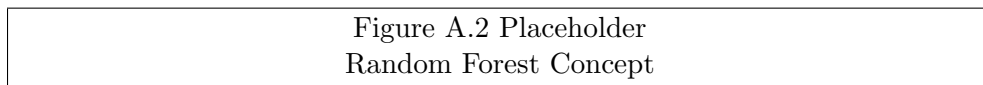


Figure 3: A conceptual illustration showing how multiple decision trees are combined to produce a single prediction.

Random forests generally achieve higher predictive performance than individual decision trees while still providing valuable information about the relative importance of predictor variables. For this reason, they have become one of the most widely used interpretable machine learning methods in educational data analysis.

A.5 Feature Importance

Many machine learning models can provide more than accurate predictions; they can also indicate which variables contributed most strongly to those predictions. This measure is commonly referred to as *feature importance*. In tree-based models, feature importance reflects the extent to which each predictor contributes to reducing classification error across all decision trees in the model. The individual predictor importances obtained in this study are shown in Figure 4, illustrating the relative contribution of each predictor to the tuned tree-based models.

Feature importance should be interpreted as a relative measure rather than an absolute effect size. The values indicate the comparative contribution of each predictor within a particular model and typically sum to one. Consequently, a feature with an importance of 0.30 contributed approximately twice as much to the model's predictive decisions as a feature with an importance of 0.15, but this does not imply a causal relationship.

In this study, feature importance was examined at two levels. Individual predictor variables were first evaluated within each model. Related variables were then aggregated into broader educational themes, including secondary mathematics preparation, previous university performance, preparatory mathematics, secondary school achievement, student characteristics, and the

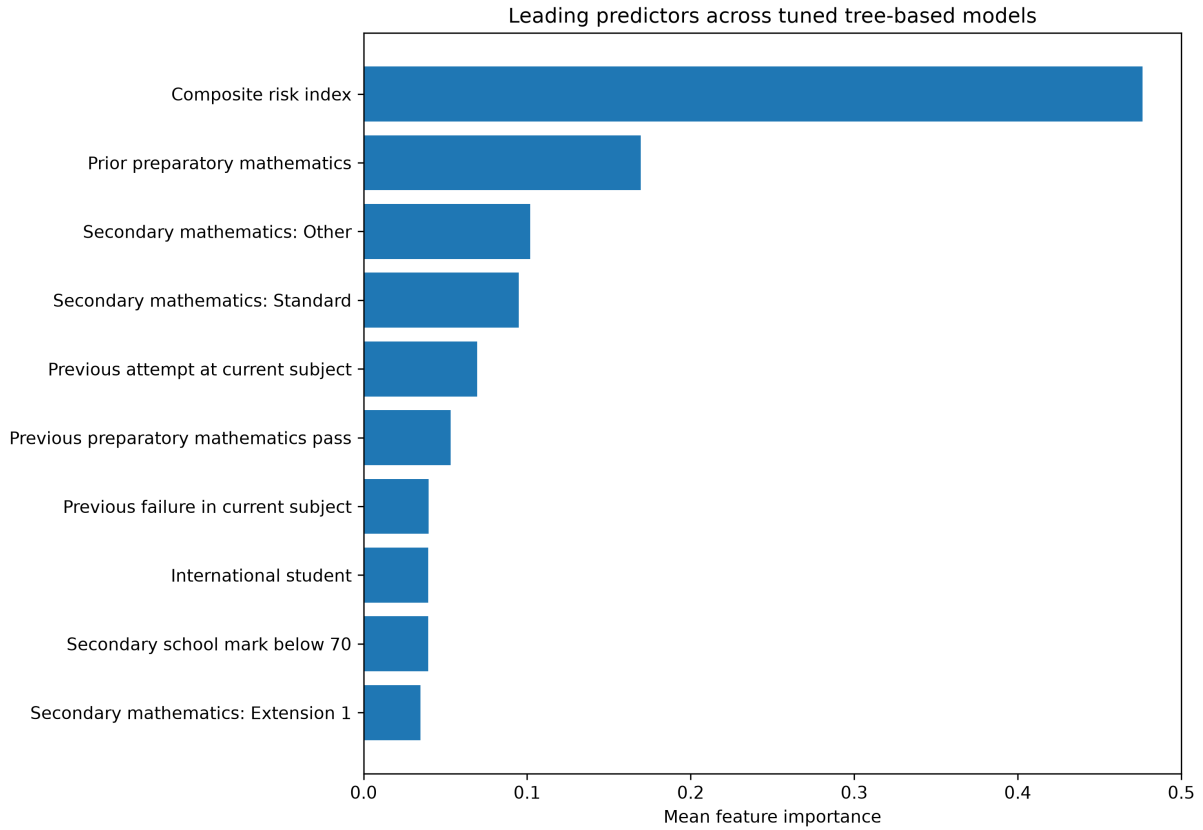


Figure 4: Individual feature importance values produced by the tuned tree-based models. Variables with larger importance contributed more strongly to the model’s predictions.

composite risk measure. This grouped interpretation provides a clearer educational perspective while preserving the underlying machine learning results.

A.6 Grouped Feature Importance

Individual feature importance values provide detailed information about the contribution of specific predictor variables. However, educational researchers are often interested in broader concepts rather than individual dummy-coded variables. For example, several variables may represent different levels of secondary mathematics preparation, while others describe previous university performance or participation in preparatory mathematics subjects.

To improve interpretability, this study grouped related predictor variables into educational themes before calculating their combined importance. This approach preserves the predictive information produced by the machine learning models while presenting the results in a form that is more closely aligned with educational theory and practice. Rather than interpreting numerous individual variables, readers can evaluate the relative importance of broader constructs that are meaningful within the context of student transition. The resulting grouped feature importance values are presented in Figure 5, providing an educational interpretation of the model outputs by aggregating related predictors into broader conceptual themes.

Grouping feature importance therefore provides an additional interpretative layer. It bridges the gap between machine learning outputs and educational understanding by translating technical model results into concepts that can be discussed within the broader literature on student preparation, transition, and academic success.

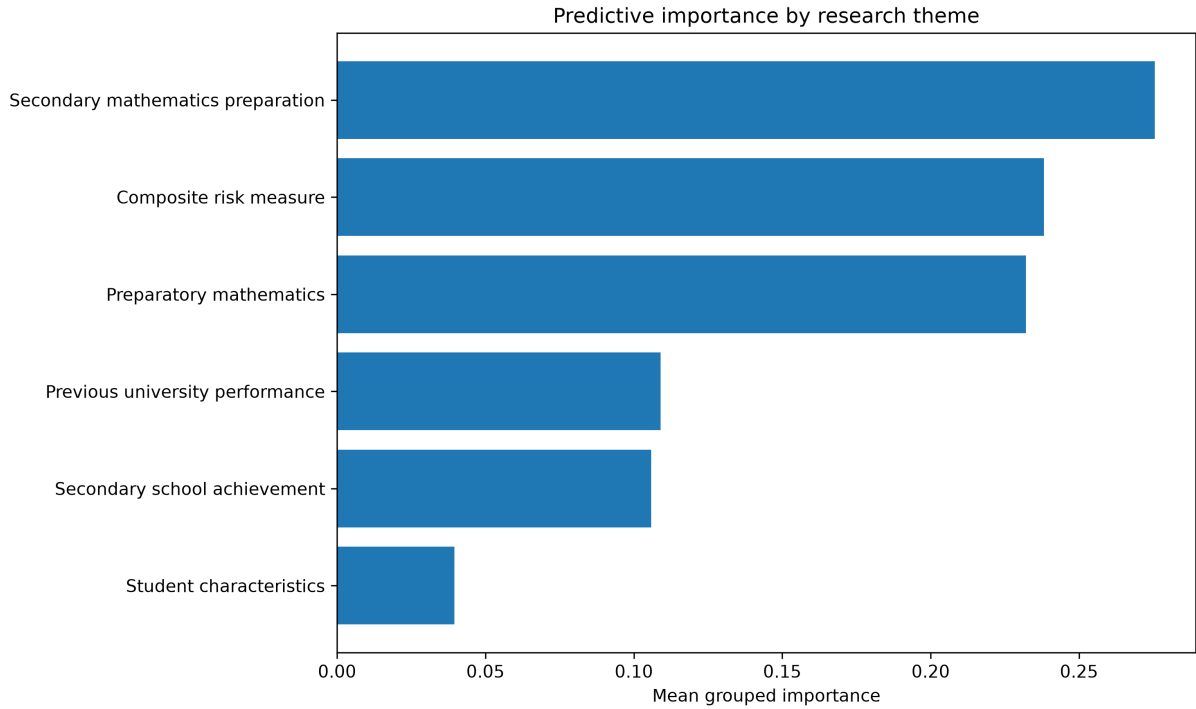


Figure 5: Grouped feature importance values showing the relative contribution of broader educational themes across the tuned tree-based models.

Part III — Model Development and Evaluation

A.7 Training and Test Data

A fundamental principle of machine learning is that model performance should be evaluated using data that were not used during model training. If a model is assessed only on the observations from which it learned, the reported performance is likely to be overly optimistic because the model has already seen those examples.

To address this issue, the available data are commonly divided into two subsets. The training dataset is used to construct and tune the model, while the test dataset is reserved for an independent evaluation of predictive performance. The test data therefore provide a more realistic indication of how well the model is expected to perform when applied to new students.

This study adopted a reproducible train–test split, ensuring that all candidate models were evaluated using the same independent test dataset. This approach provides a fair comparison between models and reduces the likelihood that reported performance reflects random variation in the training data rather than genuine predictive capability.

A.8 Cross-Validation

A single train–test split provides one estimate of model performance, but the result may depend on the particular observations assigned to each subset. Cross-validation reduces this dependence by repeatedly dividing the training data into different partitions and evaluating the model across multiple iterations.

In (k)-fold cross-validation, the training data are divided into (k) approximately equal subsets. During each iteration, one subset is reserved for validation while the remaining subsets are used for model training. This process is repeated until every subset has served as the validation set exactly once. The resulting performance measures are then averaged to provide a more stable estimate of model performance.

Cross-validation is particularly valuable during model development because it helps identify models that generalise well rather than those that perform well only for a particular sample of training data. For this reason, it is commonly used during hyperparameter tuning before a final evaluation is performed on the independent test dataset.

A.9 Hyperparameter Tuning

Many machine learning models include settings, known as hyperparameters, that influence how the model is constructed. For example, a decision tree may be limited by its maximum depth, while a random forest requires the number of trees to be specified before training begins. These values are chosen by the researcher rather than learned directly from the data.

Hyperparameter tuning is the process of systematically evaluating different combinations of these settings to identify those that produce the best predictive performance. This evaluation is commonly performed using cross-validation so that each candidate configuration is assessed consistently across multiple training and validation partitions.

Careful hyperparameter tuning improves model robustness and reduces the risk of selecting a model that performs well only because of a particular choice of settings. In this study, all candidate models were tuned before their final performance was evaluated on the independent test dataset, ensuring that comparisons were both fair and reproducible.

A.10 ROC Curves and Model Evaluation

Receiver Operating Characteristic (ROC) curves provide a standard method for evaluating the discriminatory performance of binary classification models. Rather than assessing a model using a single classification threshold, an ROC curve summarises performance across all possible thresholds by plotting the true positive rate (sensitivity) against the false positive rate ($1 - \text{specificity}$). This provides a threshold-independent assessment of how effectively a model distinguishes between the two outcome classes.

The overall discriminatory ability of a model is commonly summarised by the Area Under the ROC Curve (AUC). An AUC value of 0.5 indicates performance equivalent to random guessing, whereas a value of 1.0 represents perfect discrimination. In educational prediction studies, the ROC curve therefore provides a useful complement to measures such as accuracy, precision, and recall because it evaluates the model's ability to rank students according to their likelihood of success or failure, independent of any particular decision threshold. The ROC curves for the tuned decision tree and random forest models are presented in Figure 6, allowing their discriminatory performance to be compared across all classification thresholds.

In this study, ROC curves were used to compare the predictive performance of the tuned decision tree and random forest models using the independent test dataset. This approach enabled the relative discriminatory performance of the competing models to be evaluated consistently and provided an objective basis for selecting the best-performing model while reducing the influence of any single classification threshold.

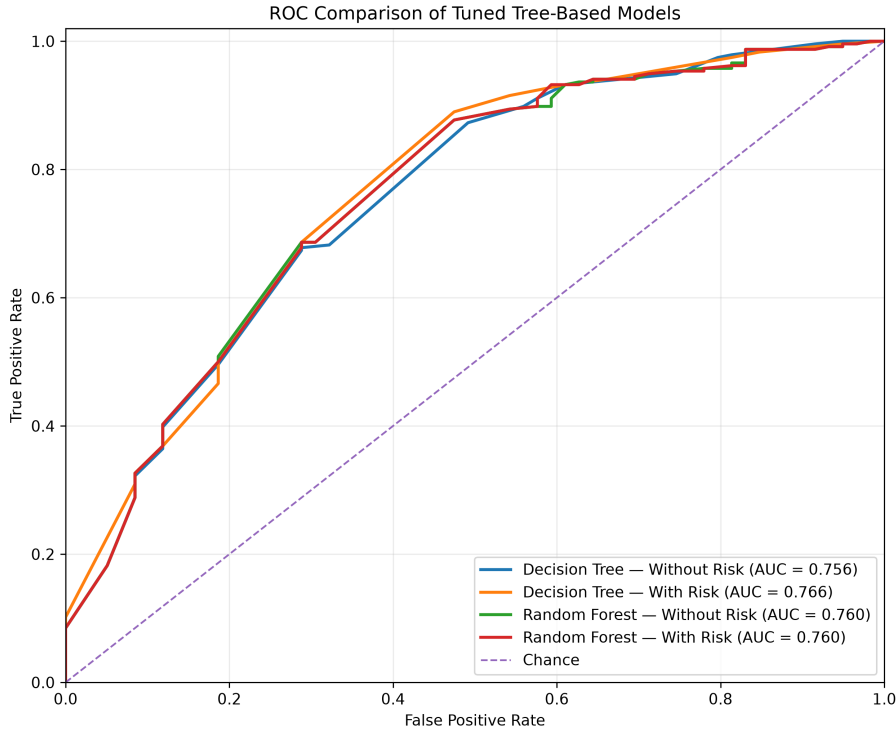


Figure 6: Receiver Operating Characteristic (ROC) curves comparing the discrimination performance of the tuned tree-based models on the independent test dataset. Curves closer to the upper-left corner indicate stronger discriminatory performance, while the area under each curve (AUC) summarises overall model effectiveness across all classification thresholds.

Part IV — Research Methodology

A.11 Why Statistical Models and Machine Learning Were Both Used

Statistical modelling and machine learning provide complementary perspectives rather than competing methodologies. Statistical models are particularly valuable when the objective is to estimate relationships between variables, test hypotheses, and quantify uncertainty. Machine learning methods are designed to maximise predictive performance and to identify complex patterns that may not be apparent using traditional statistical techniques alone.

This study deliberately combined both approaches. Logistic regression provided a well-established statistical benchmark against which the machine learning models could be compared. Decision trees and random forests were then used to investigate predictive performance and to examine the relative importance of educational factors influencing student success.

Using both approaches strengthened the overall analysis. Agreement between the statistical and machine learning results increased confidence in the findings, while differences between the models provided additional insight into the strengths and limitations of each analytical approach.

A.12 Reproducible Computational Workflows

Reproducibility is a fundamental principle of modern computational research. A reproducible workflow ensures that the complete analysis, from data preparation to the final manuscript, can be repeated using the same source data and documented procedures. This reduces the likelihood of transcription errors, improves transparency, and enables other researchers to verify and extend the reported findings.

In this study, reproducibility was supported through an integrated computational workflow. Data preparation, statistical modelling, machine learning analyses, independent verification,

figure generation, table production, and manuscript components were all produced automatically from the underlying analysis. The paper itself incorporated generated values directly from the analysis pipeline, ensuring consistency between reported results and computational outputs. The overall computational workflow developed for this research is summarised in Figure 7, illustrating how the study progresses from raw research data through automated analysis, independent verification, and manuscript generation.

Such workflows provide benefits beyond a single study. They simplify maintenance, facilitate future analyses on new datasets, and support transparent, evidence-based educational research by ensuring that published results remain directly traceable to the underlying computations.

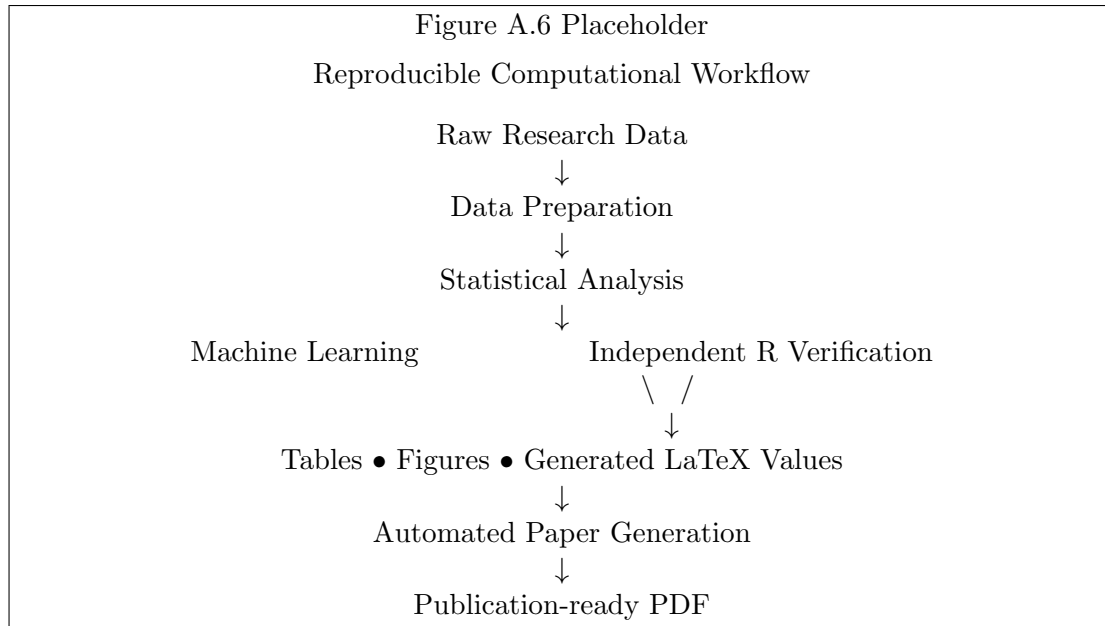


Figure 7: Overview of the reproducible computational workflow developed for this study, illustrating the progression from raw research data through analysis, independent verification, automated reporting, and final manuscript generation.

References

A Supplementary Material